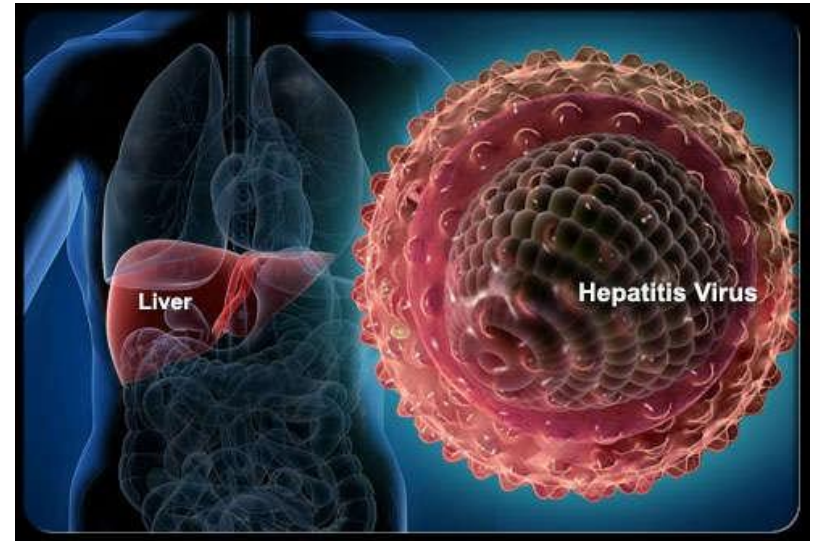


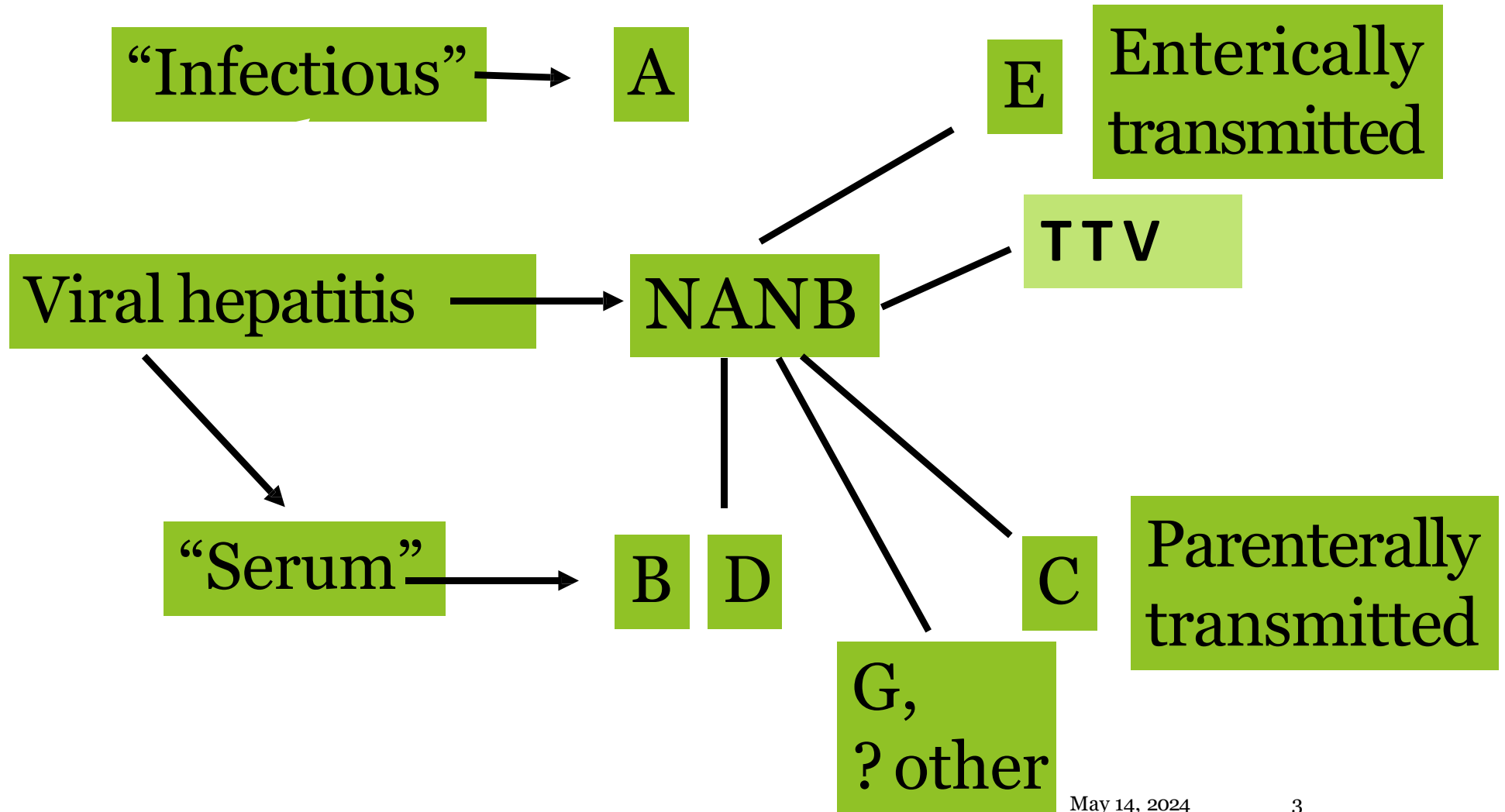
Viral Hepatitis



Introduction

- ▶ **Hepatitis:** inflammation of liver; presence of inflammatory cells in organ tissue
- ▶ The term **VIRAL HEPATITIS** is usually used to describe infections caused by viral agents whose primary tissue tropism is the liver
- ▶ More than 90% of viral hepatic infections are caused by five hepatitis virus designated as from A-E:
 - ▶ Hepatitis A → Epidemic hepatitis
 - ▶ Hepatitis B → Serum hepatitis (injection hepatitis)
 - ▶ **Hepatitis C → Parenterally transmitted non-A, non-B hepatitis**
 - ▶ Hepatitis D → Delta hepatitis
 - ▶ Hepatitis E → Enterically transmitted non-A, non-B hepatitis
- ▶ However, also less well known hepatitis G and TTV

Viral Hepatitis – Historical Perspective



- ▶ These hepatitis viruses belong to different families;
 - ▶ distinct from each other in
 - ▶ morphology,
 - ▶ physicochemical properties,
 - ▶ biology and
 - ▶ epidemiology
- ▶ infect primarily the liver, causing inflammation and hepatocellular necroses
- ▶ The majority of infections are totally **asymptomatic**
- ▶ Clinical symptoms ➔ **similar in all acute hepatitis infections:**
 - ▶ Incubation period => characteristic of the specific agent
 - ▶ Early symptoms (prodromal phase) persist for several days

Patterns of hepatitis

- ▶ **Carrier state:**
 - ▶ is an individual who harbor and can transmit the virus but has no manifest symptoms.
- ▶ **Acute hepatitis:**
 - ▶ if its manifestation persist for period less than six months.
- ▶ **Chronic hepatitis:**
 - ▶ if there is clinical or serological evidence of liver pathology persistent for more than six consecutive months.
- ▶ **Fulminant hepatitis:**
 - ▶ if massive hepatic cell necrosis happened within few weeks leading to acute hepatic failure and hepatic encephalopathy.
- ▶ **Cirrhosis:** Replacement of liver tissue → fibrosis, scar tissue

Mechanism of Injury

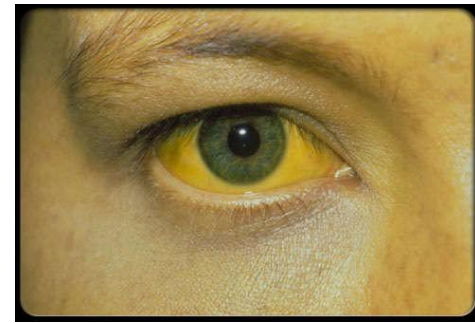
- **Acute infection:**
direct cytotoxicity **VS** immune response to infection **VS** combination of the two
- **Chronic infection:**
immune response probably more important

Clinical Manifestations of Acute Viral Hepatitis

- ▶ Phase 1 - Viral replication; Patients are asymptomatic during this phase.
- ▶ Phase 2 – Prodromal:
 - **Early symptoms** persist **for several days**
 - Fever, extreme physical or mental tiredness (exhaustion or fatigue), loss of appetite (anorexia), nausea and vomiting, lose of sense of smell and taste
- ▶ Phase 3 - Icteric phase
- ▶ Phase 4 - Convalescent phase;
 - ▶ symptoms and icterus resolve.
 - ▶ Liver enzymes return to normal.

► Icteric phase:

- phase relating to the appearance of jaundice which develop late
- starts 3-5 days after prodromal phase:
 - Dark urine, pale colored stool, jaundice (this is the hall mark of hepatitis infection)
- *raised liver enzymes AST (aspartate aminotransferase)*
 - *this is an enzyme that catalyzes the reversible transfer of an amine group from L-glutamic acid to oxaloacetic acid, (forming - ketoglutaric acid and L-aspartic acid) and*
- *ALT (Alanine aminotransferase)*
 - *this is an enzyme that transfers amino groups from L-alanine to 2-ketoglutarate, or the reverse (from L-glutamate to pyruvate).*



Asymptomatic



Symptomatic

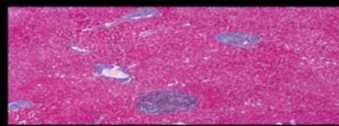
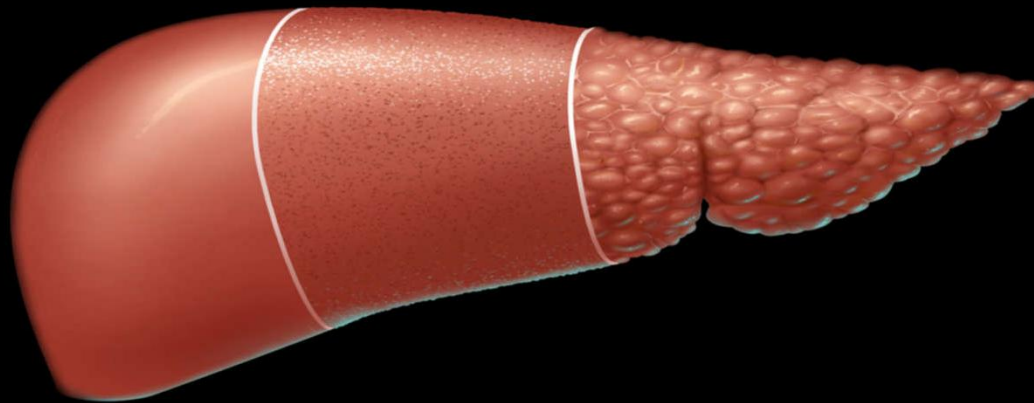


**Fulminant liver failure
and death**

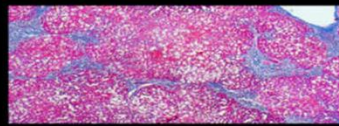
Typical Signs and Symptoms

- **Fever**
- **Malaise**
- **Loss of appetite (Anorexia)**
- **Nausea**
- **Vomiting**
- **Jaundice**
- **Scleral icterus**
- **Abdominal/Right Upper Quadrant(RUQ) pain**
- **Hepatomegaly**
- **Arthritis**
- **elevation of ALT, AST, special hepatitis markers**

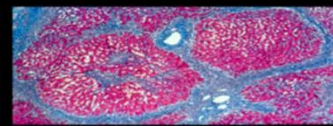
Long-Term Sequelae of Chronic Hepatitis



Normal liver



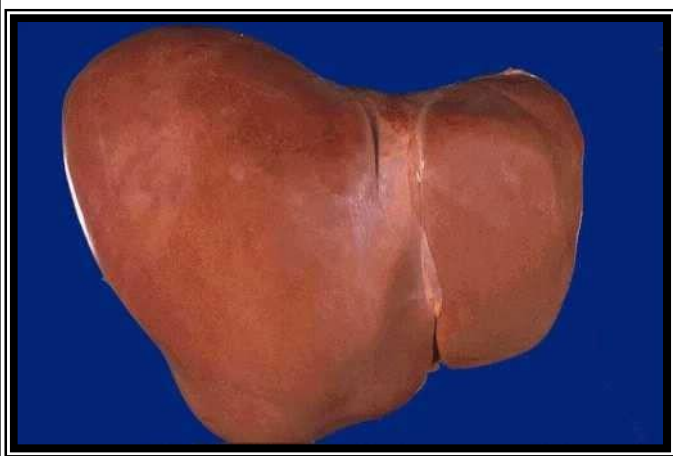
Chronic hepatitis



Cirrhosis

Hepatocellular
carcinoma

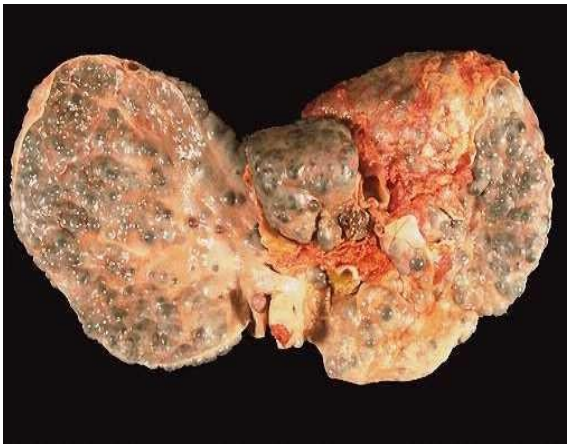
Normal Liver



Micronodular cirrhosis



Macronodular Cirrhosis



Specific diagnosis:- serology

- Presence of Viral Proteins (mostly called 'antigens') /Nucleic acid
 - Virus is present
 - Virus might be replicating
- Presence of antibodies to Viral proteins
 - Virus may be currently present (or not)
 - Could indicate either immunity or ongoing infection

Immunologic Markers Used for Serologic Diagnosis

Immune globulin of the M subclass (IgM):

- **Produced in early infection**

Immune globulin of the G subclass (IgG):

- **Produced in later infection**

Total immune globulin (Total Ig):

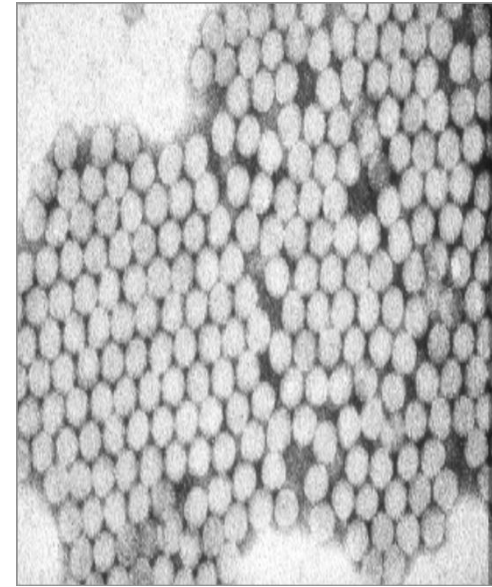
- Combination of IgM and IgG
 - Early infection - primarily IgM
 - Late infection - primarily IgG

Hepatitis A virus (HAV)

The causative agent of infectious hepatitis

The virus

- ▶ Belongs to the family Picornaviridae
- ▶ Genus *Enterovirus*:
 - ▶ Non-enveloped, small virion
 - ▶ ss +sense RNA (7.5)
 - ▶ Surrounded by capsid containing three structural proteins (VP1, VP2, and VP3)



▶ **HAV is very hardy:**

- ▶ **Tolerates pH** as low as 3.0 for more than an hour
- ▶ **Survives heat** inactivation at 60°C for an hour
- ▶ **Maintains its replicative capacity** at room temperature **for** several weeks
- ▶ **Not fully inactivated by 70% ethanol** disinfection at room temperature;
 - ▶ **needs formalin treatment** (3% for 5 min at room temp) or **boiling** for 1 min

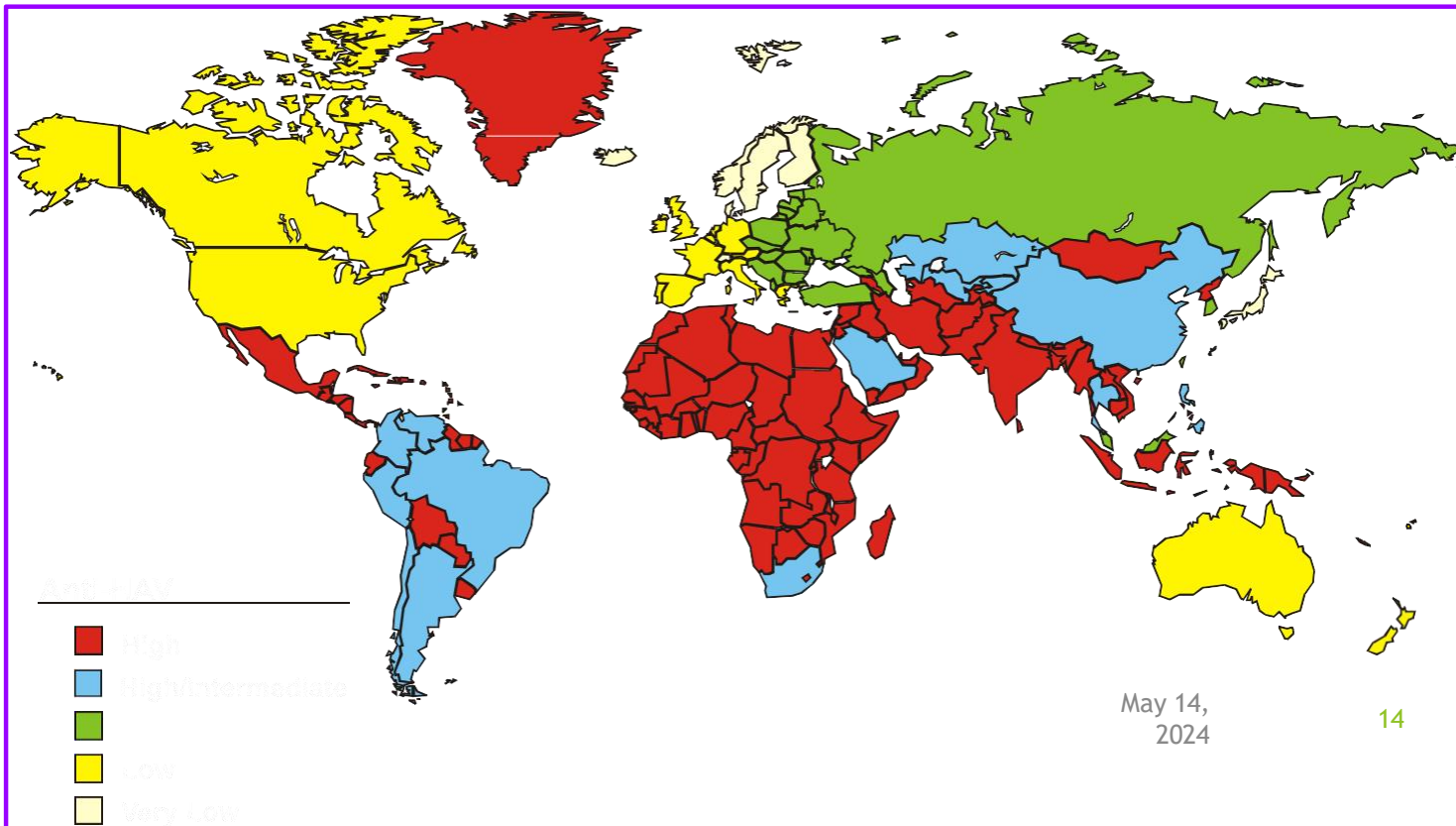
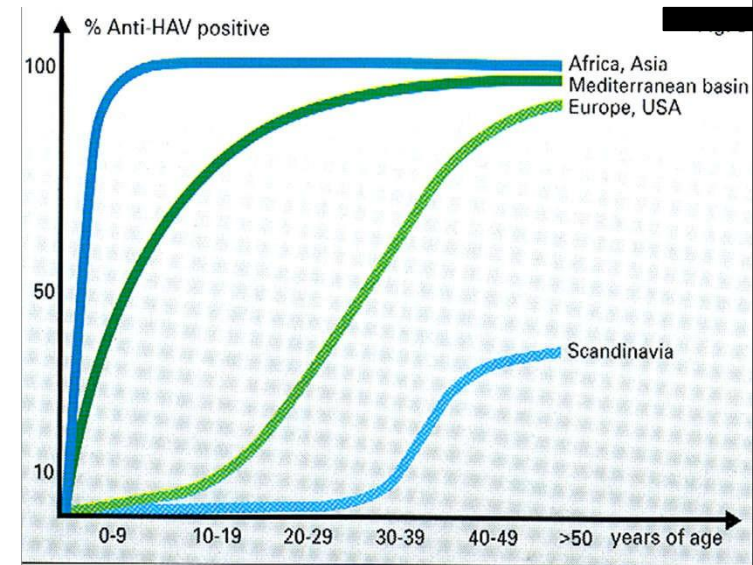
Epidemiology

▶ Route of transmission:

- ▶ Fecal-oral route: virus ingested orally
 - ▶ Direct contact with infected individuals (most frequent)
 - ▶ Drinking fecally-contaminated water Or
 - ▶ Eating contaminated food
- ▶ Large quantities of **viral particles excreted in stools** → **peak level** being during the **incubation period (15-49 days)** just before onset of clinical syndromes
- ▶ **Parenteral transmission occurs, BUT it is rare** (epidemiologically insignificant)

Spread:

- Widely distributed
- Nearly 100% of children infected by the age of 5
- Poor hygiene is risk factor



Pathogenesis

- ▶ Virus enters via the gut → replicates in the alimentary tract → spreads to infect the liver (where it multiplies in hepatocytes)
- ▶ Viremia is transient
- ▶ Virus is excreted in the stools for two weeks preceding the onset of symptoms
- ▶ Events involved in the pathogenesis of liver injury after natural infection with HAV is poorly understood
- ▶ Virus is not cythopathic
- ▶ However,
 - ▶ host immune cells such as natural killer cells or cytolytic T lymphocytes rather than the virus may mediate alteration of the liver tissue

Clinical features

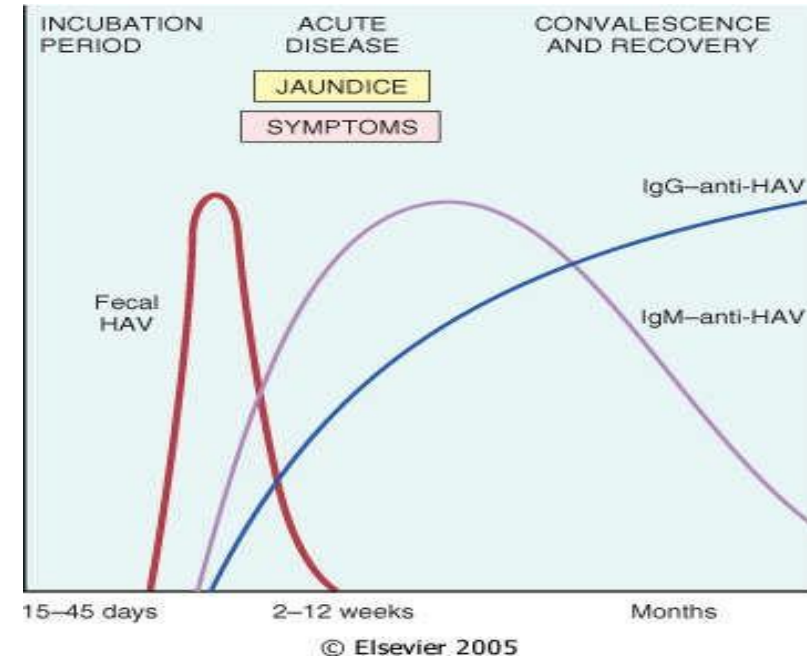
- ▶ Incubation period 3-5 weeks (mean 28 days)
- ▶ Clinical symptoms **depend on age**:
 - ▶ In **90-95%** of children **under 5**, it **proceeds without icterus or any clinical signs**
 - ▶ **2/3 of teenagers** and **adults** experience an **icteric course**
- ▶ About **0.1% of all clinically manifested acute hepatitis A** become **Fulminant** (rapidity and great intensity or severity) with **frequent FATAL OUTCOME**
- ▶ Fulminant disease increases with age

In general, except few fatal cases:

- HAV patients recover fully (self-limiting)
- **Chronic** infections are **not known**;
 - 5-20% last long time → after initial symptoms subsided temporarily,
- new disease episodes may occur → after 6-12 months these also disappear

Diagnosis

- ▶ Detection of HAV-specific antibodies in serum
 - ▶ **IgM anti-HAV**: appears 4 wks after exposure and disappears by 3 -6 months. Indicates **acute** infection
 - ▶ **IgG anti-HAV**: peaks during convalescence and persists for life → Indicates **exposure and immunity**
- ▶ Detection of Virus in stools (PCR)
- ▶ Detection of HAV antigens in stool



Therapy and prevention

- No specific therapy for acute HA
 - Supportive therapy.
 - Immunity after HAV infection is usually lifelong

Prevention: → Proper hygiene

Prevention of Hepatitis A with IG (passive immunization)

Pre-Exposure Prophylaxis

- Protection lasts 3-5 months
- Used for travelers

Post-Exposure Prophylaxis

- Must be given within 14 days of exposure
- Used for household and sexual exposures
- Sometimes used for common source exposures (e.g., infected food handler)

Prevention of Hepatitis A with Hepatitis A Vaccine

▶
▶
1st single injection in the arm gives protection for a year.

2nd booster injection at 6 to 12 months extends protection for at least 20 years

▶ Licensed for use for persons

- Enterically transmitted
- non-A, non-B (NANB) hepatitis

The Virus

- ▶ HEV is a small (32nm)
- ▶ Non-enveloped
- ▶ RNA virus

- ▶ Originally classified under Caliciviridae
 - ▶ some claim that molecular studies suggest it is related more to rubella virus (togavirus) although it is non-enveloped.

Epidemiology

- **Transmission:** ... fecal-oral (fecally contaminated drinking water more common source)
- **Spread:** Present in **animals without causing disease**
 - Socioeconomically disfavored regions of the world
 - Endemic in Tropical and Subtropical countries

Clinical features

- Incubation period ... **about 40 days**
- Manifested as acute viral hepatitis A, but **more severe than HAV**;
 - but self-limiting; recovery without complication
- Mortality = 1%; **Fulminant hepatic necrosis in pregnant women (case fatality rate is 10-50%)**
- **No chronicity**

Diagnosis

- IgM antibodies to HEV,
- HEV RNA assay

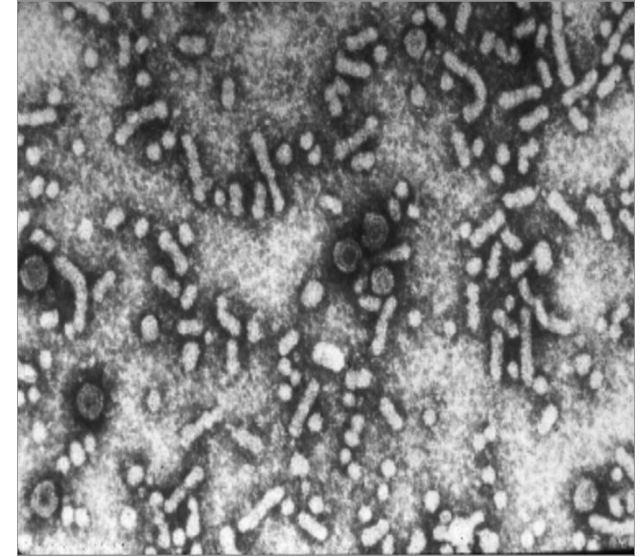
• Therapy and immunoprophylaxis

- **No specific therapy;**
- **no vaccine**

Hepatitis B Virus (HBV)

The Virus

- ▶ Belongs to the family Hepadnaviridae
- ▶ Enveloped virus
- ▶ **Three important antigens:**
 - ▶ HBsAg → surface (coat) protein
 - ▶ three forms – small, medium, and large
 - ▶ produced in excess as small spheres and tubules
 - ▶ HBcAg → inner core protein
 - ▶ HBeAg → secreted protein; function unknown
- ▶ **Genome circular partially double stranded DNA (the + sense strand shorter)**



- ▶ Serum from HBV patient contains three morphologically distinct particles:

- ▶ the **Dane particles**

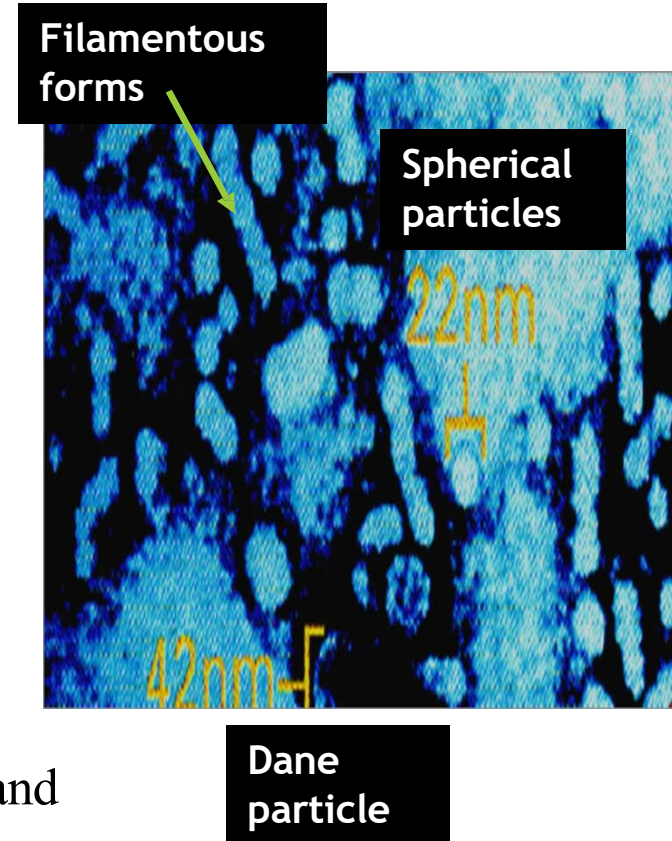
- ▶ contains every structure found in a **complete HBV virus**:
 - ▶ lipoprotein coat with surface antigens,
 - ▶ the viral capsid core antigen in which
 - ▶ viral genome and
 - ▶ viral polymerase are enclosed;

- ▶ the **Spherical particles (22nm)**

- ▶ which contain **only the lipoprotein coat antigens** and

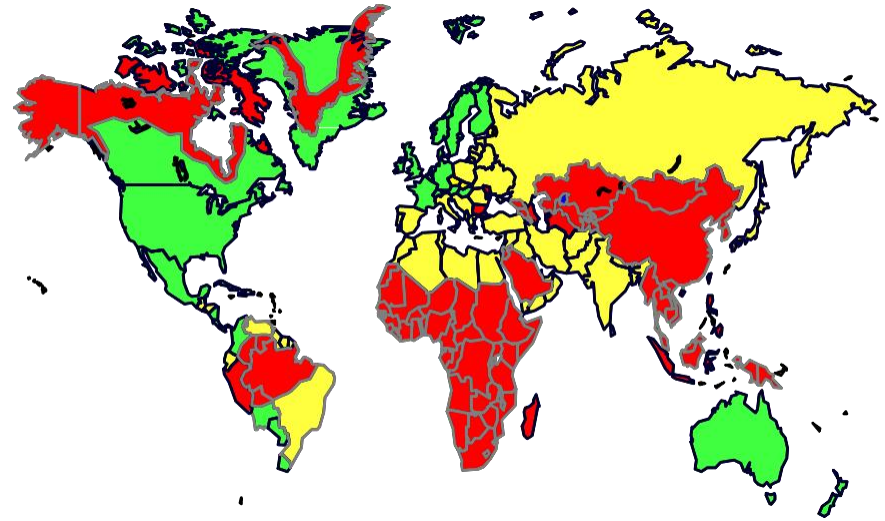
- ▶ the **Filamentous forms**,

- ▶ **which have the same diameter and lipoprotein coat** just like the Spherical ones



Epidemiology

- ▶ **Routes of transmission:**
 - ▶ Parenteral transmission
 - ▶ **Body fluids such as serum**, saliva, and semen are found to contain infectious particles
 - ▶ **Blood transfusion**, organ donation, renal dialysis
 - ▶ **Direct inoculation** (contaminated needles, scalpels, nail clippers, shavers, etc)
 - ▶ **Microlesions of mucus membranes** (e.g. via **sexual contact**; (usually anal sex for MSM people)
 - ▶ **Transmission from carrier mother to neonate** (before, during and after delivery) = **vertical transmission**



HBsAg Prevalence

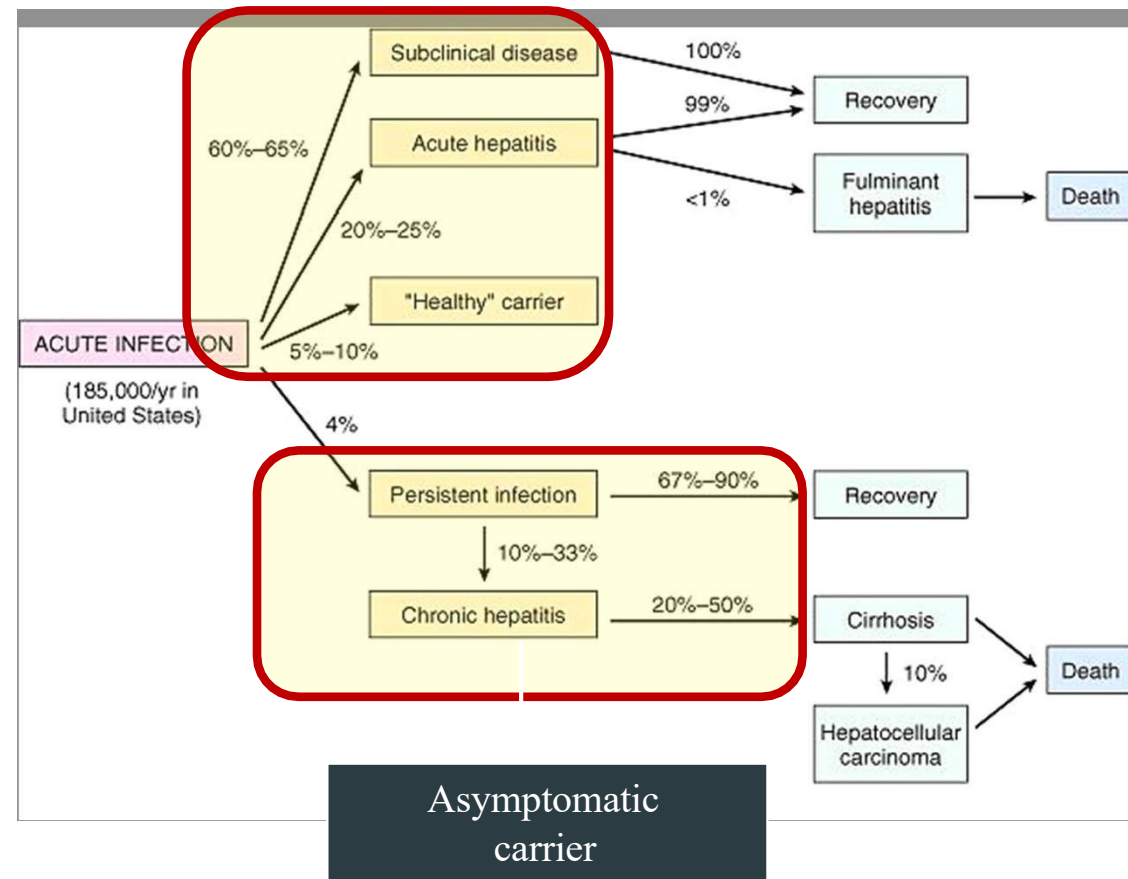
- $\geq 8\%$ - High
- 2-7% - Intermediate
- $< 2\%$ - Low

Clinical features:

- ▶ Incubation period 2 - 5 months

Acute infection:

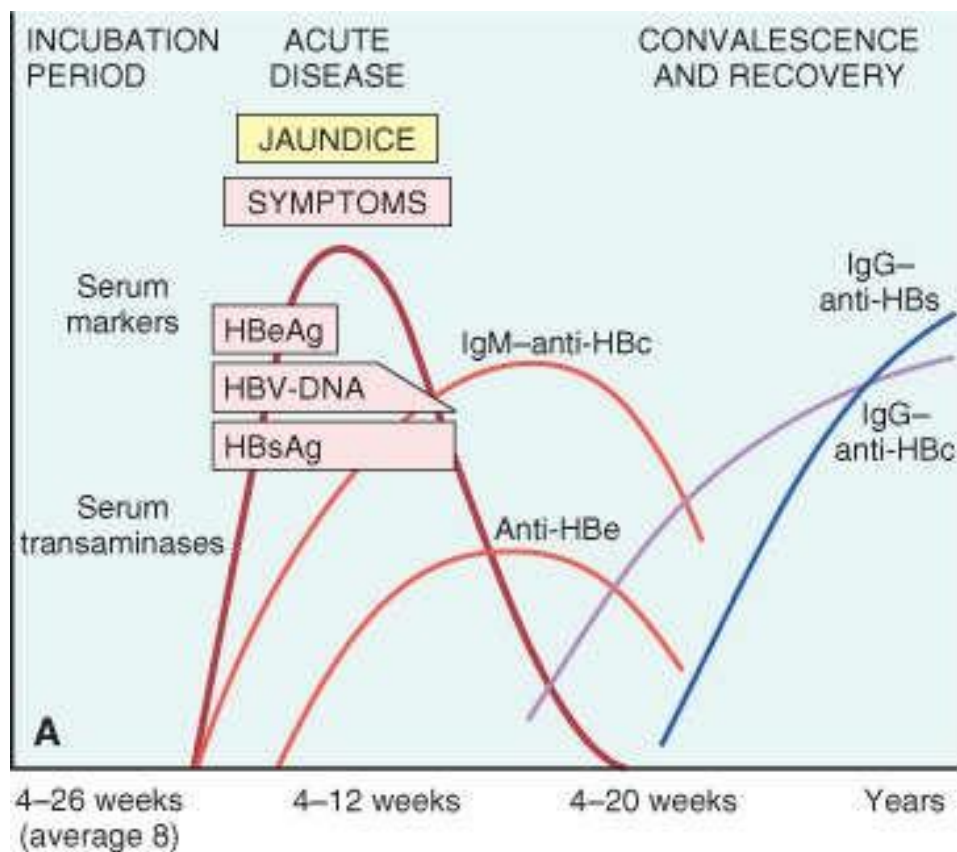
- ▶ Gradual but harmful onset
 - ▶ The majority of acutely infected patients do not develop clinical symptoms
 - ▶ Only about 1% develop fulminant hepatitis
- ▶ The rest develop mild clinical symptoms:



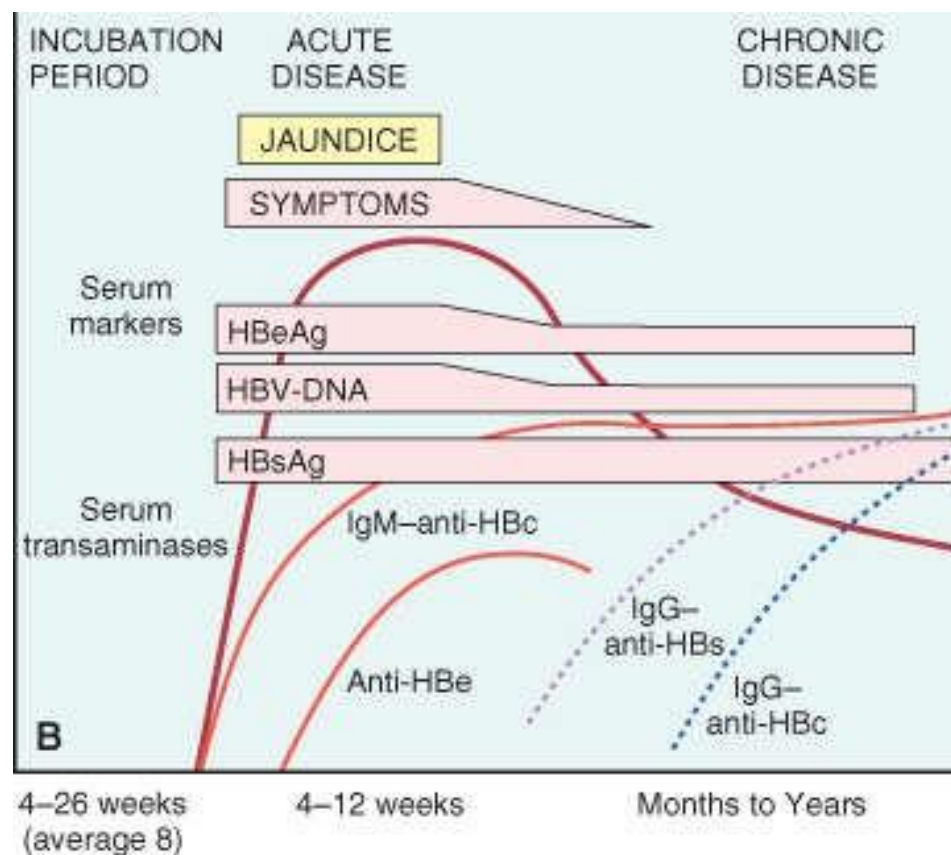
- ▶ **Symptoms are often non-specific symptoms** such as myalgia, malaise, nausea, fatigue, pruritus, abdominal pain, RUQ, jaundice
- ▶ Fulminant Hepatitis--Acute HBV results in Liver Failure

Acute	Chronic
May be asymptomatic	Usually asymptomatic
Flu-like symptoms	Malaise/Fatigue
Skin Rash	Extra-hepatic symptoms
Jaundice	Signs/symptoms of liver failure
Light-colored stools	Hepatocellular carcinoma
Dark-colored urine	

Viral Hepatitis B: Serology



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Sequence of serologic markers for hepatitis B viral hepatitis demonstrating

(A) acute infection with resolution

(B) progression to chronic infection.

Pathogenesis of HBV infection

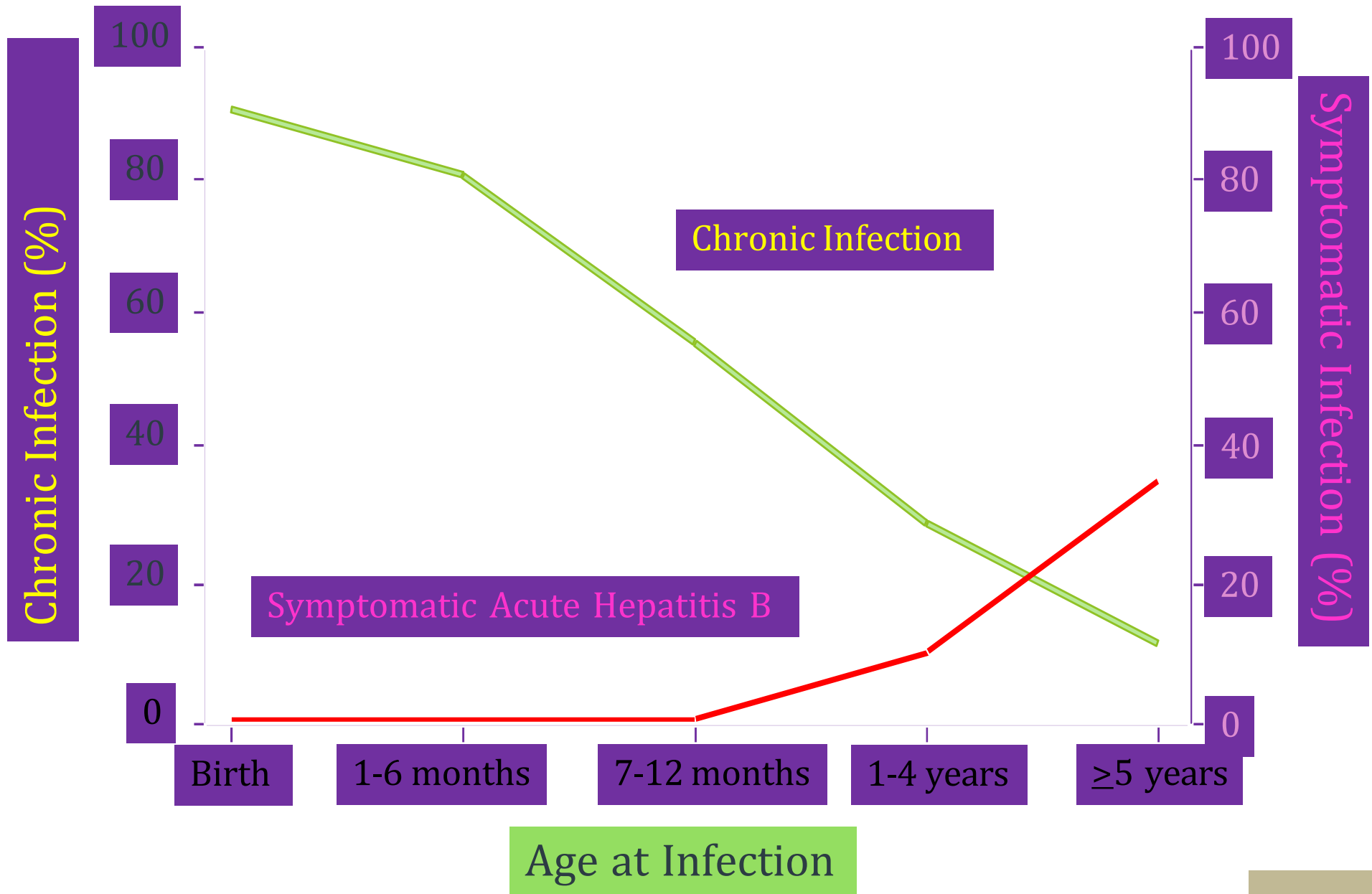
- ▶ The virus replicates in the liver
- ▶ Virus particles as well as excess viral surface protein are shed in large amounts into the blood
- ▶ Viremia is prolonged and the **blood of infected individuals is highly infectious**

▶ Acute infection:

- ▶ Mechanism of liver cell injury during infection not well understood
1. Host immune system, particularly the cellular immunity (not the humoral immunity) suspected to be involved:
 - ▶ CTL destruction of infected liver cells
 - ▶ **Indirect evidences:**
 - ▶ Infections at very young ages are associated with very mild initial hepatitis
 - ▶ Immunologically impaired patients develop milder initial disease than immunologically competent patients

2. Expression of viral antigens (such as HBcAg and HBsAg) and their accumulation in the infected cells → toxic to the cells

Outcome of HBV Infection by Age at Infection



Persistent or chronic infection:

- ▶ The virus persists in the hepatocytes and **on-going liver damage** occurs because of the host immune response against the infected liver cells
- ▶ **Chronic infection** may take one of two forms:
 - ▶ *Chronic persistent Hepatitis* - the virus persists, but there is minimal liver damage
 - ▶ *Chronic Active Hepatitis* - There is aggressive destruction of liver tissue and rapid progression to cirrhosis
 - ▶ end stage liver disease characterized by diffuse damage to hepatic parenchymal cells, with nodular regeneration, fibrosis, and disturbance of normal architecture (Cirrhosis) or liver failure
 - ▶ Patients who become persistently infected are at risk of developing **hepatocellular carcinoma (HCC)**



Diagnosis of HBV

Detection of virus antigens:
HBsAg, HBeAg

Detection of Antibodies:
Anti-HBs, anti-HBc, anti-HBe

Detection of viral nucleic acid (DNA)

Concentration of HBV in Body Fluids

High

blood

serum

wound exudate

Moderate

semen

vaginal fluid

saliva

**Low/not
Detectable**

urine

feces

sweat

tears

breast milk

HBsAg

- ▶ First specific marker
- ▶ Detectable during incubation, peaks in acute stage
- ▶ Declines upon recovery
- ▶ Elevated in carriers
- ▶ Screening test for donor blood

HBeAg

- ▶ Appears shortly after HBsAg and parallels HBsAg
- ▶ Present **during active replication** of virus (most infectious phase)

Anti-HBc

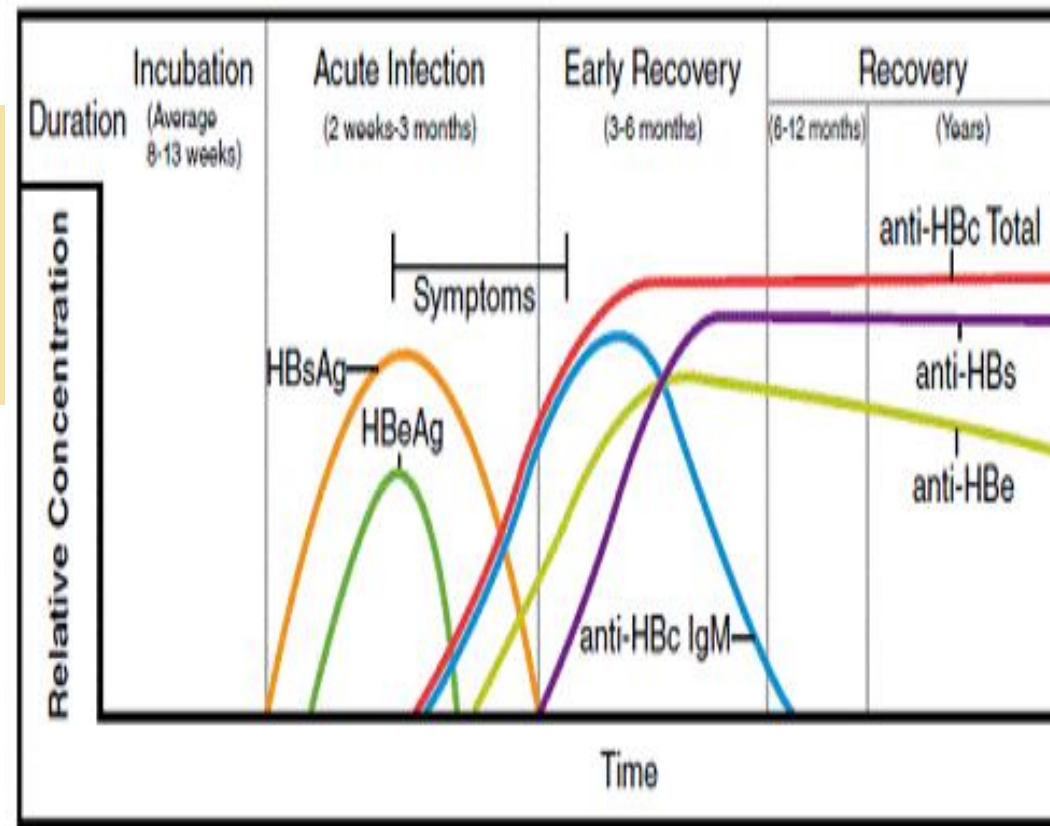
- ▶ First detectable antibody
- ▶ IgM **present** in the interval between disappearance of HBsAg **and** appearance of Anti-HBs (**core window**)
- ▶ IgG produced during convalescence and persists for life

Anti-HBe

- ▶ Appears after disappearance of HBeAg
- ▶ Indicates resolution

Anti-HBs

- ▶ Appears during recovery and lasts for years
- ▶ Indicates immunity (also produced as a result of vaccination)



Interpretation of Tests for Acute Hepatitis B

Anti-HBc IgM	Anti-HBc IgG	HBsAg	Anti-HBs	Interpretation
Positive	Negative	Positive	Negative	Acute HBV infection
Negative	Negative	Positive	Negative	Early acute HBV infection
Negative	Positive	Negative	Positive	Resolved acute HBV infection
Negative	Negative	Negative	Positive	Not infected Prior vaccination for HBV
Negative	Negative	Negative	Negative	Not infected
Negative	Positive	Positive	Negative	Chronic HBV infection

Hepatitis B Immune Globulin (HBIG)

- Post-exposure prophylaxis
 - ▶ Prepared from plasma containing high titers anti-HBs
 - ▶ Used for post-exposure prophylaxis to prevent infection from:
 - Perinatal exposure
 - Sexual exposure
 - Occupational exposure
 - ▶ Must be given soon after exposure to be effective
 - ▶ Does not protect against future exposures

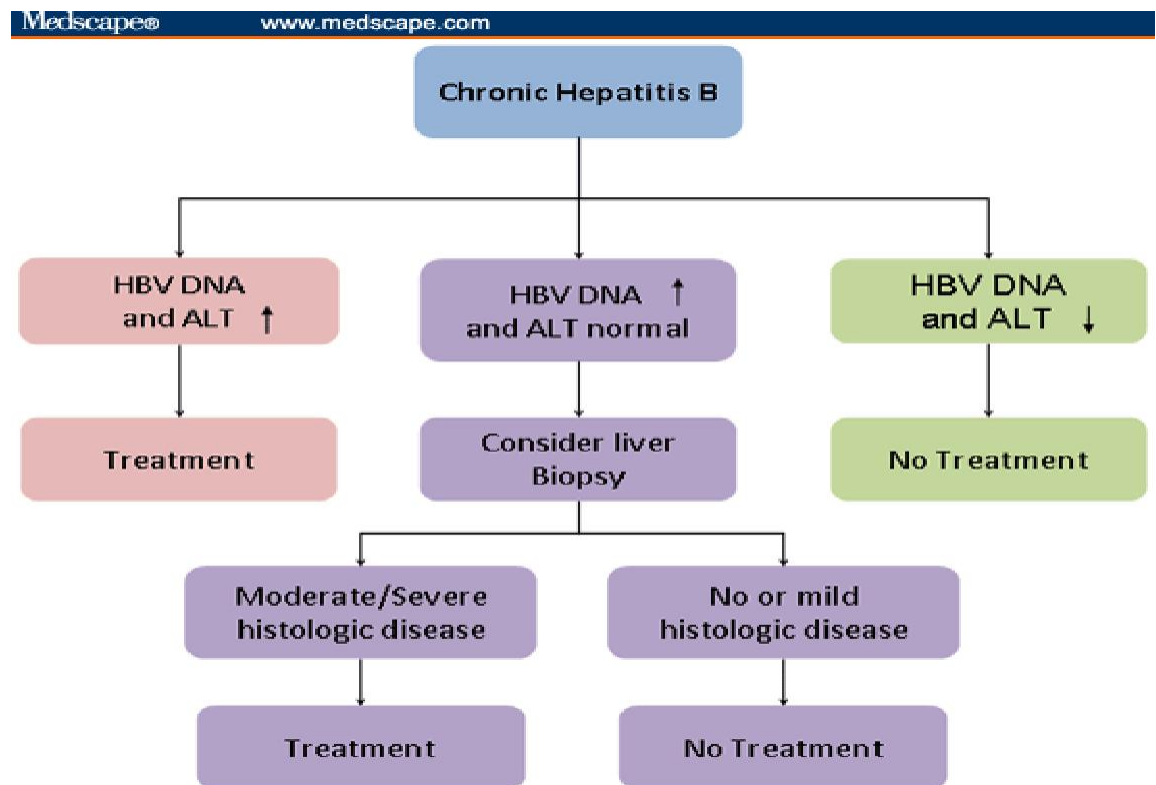
Hepatitis B Vaccine

-
-
- ▶ Primary component of vaccine is HBsAg
 - ▶ HBsAg elicits development of anti-HBs
 - ▶ anti-HBs protective antibody
- ▶ Worldwide, plasma-derived and recombinant formulations available
- ▶ **Given as a 3-dose series**
- ▶ Highly immunogenic – seroconversion ~95%
- ▶ Protection long-lasting

Treatment

- ▶ Acute hepatitis B infection does not usually require treatment because most adults clear the infection spontaneously
- ▶ On the other hand, treatment of chronic infection may be necessary to reduce the risk of cirrhosis and liver cancer.
- ▶ Although none of the available drugs can clear the infection, they can stop the virus from replicating, thus minimizing liver damage.

Treatment Algorithm



- **Interferon therapy** (immune system modulators) – First Line
 - alpha-2a and
 - PEGylated (**Polyethylene glycol**) interferon alpha-2a (Pegasys).
 - for HBeAg +ve carriers with chronic active hepatitis.
 - Response rate is 30 to 40%.

Antiviral drugs

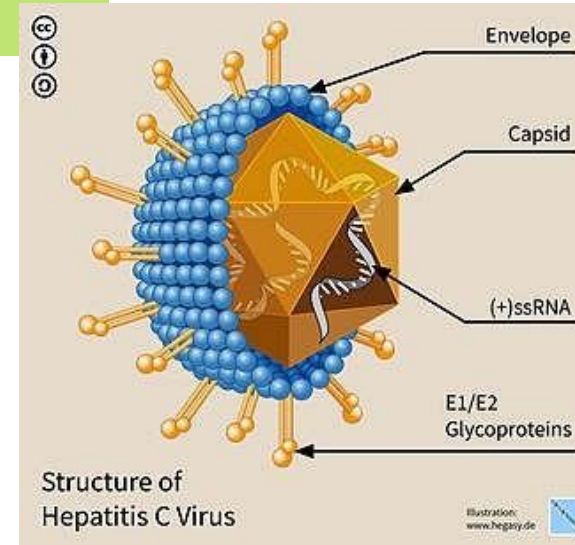
- ▶ **Lamivudine (Epivir)**: a nucleoside analogue reverse transcriptase inhibitor. Well tolerated, most patients will respond favorably. However, **tendency to relapse on cessation of treatment**. Another problem is the rapid **emergence of drug resistance**
- ▶ **Adefovir (Hepsera)**: **less likely** to develop **resistance** than Lamivudine and may be used to treat Lamivudine resistance HBV. However **more expensive** and **toxic**
- ▶ **Tenofovir (Viread)**: Nucleotide analogue
 - Highly effective with low resistance
 - Well tolerated
- ▶ **Telbivudine (Tyzeka) and entecavir (Baraclude)**:
 - ▶ Worse resistant profile than Entecavir
 - ▶ Side effect: lactic acidosis
- **Entecavir** – most powerful antiviral known, similar to Adefovir
 - Successful response to treatment will result in the disappearance of HBsAg, HBV-DNA, and seroconversion to HBeAg

May 14
2024

The Virus

Hepatitis C Virus (HCV)

- ▶ Also known as parenterally-transmitted non-A, non-B hepatitis
- ▶ Family Flaviviridae
- ▶ Genotypically divergent group: there are about 6 **genotypes** (numbered from 1 to 6) and 11 subtypes identified so far
 - ▶ Association between mode of transmission and genotype: type 3 more prevalent in iv drug users
- ▶ The virus is enveloped, spherical icosahedral with a diameter of 55-65nm
- ▶ Genome is single stranded + sense RNA with length of 9.5 kb



► Route of transmission

► Parenterally:

- Blood transfusion (2.1%),
 - needle sharing in I.V. drug users (48.3%),
 - professional risk or healthcare workers (1.8%),
 - hemodialysis (0.5%)

► From HCV carrier mother →

- depends on the concentration of the virus in the blood
 - rare at low maternal virus level, i.e. $<10^6$ virus particle /mL blood
 - At high maternal virus level ($>10^6$ particles /mL of blood), the risk of the neonate to acquire the disease is about 50%

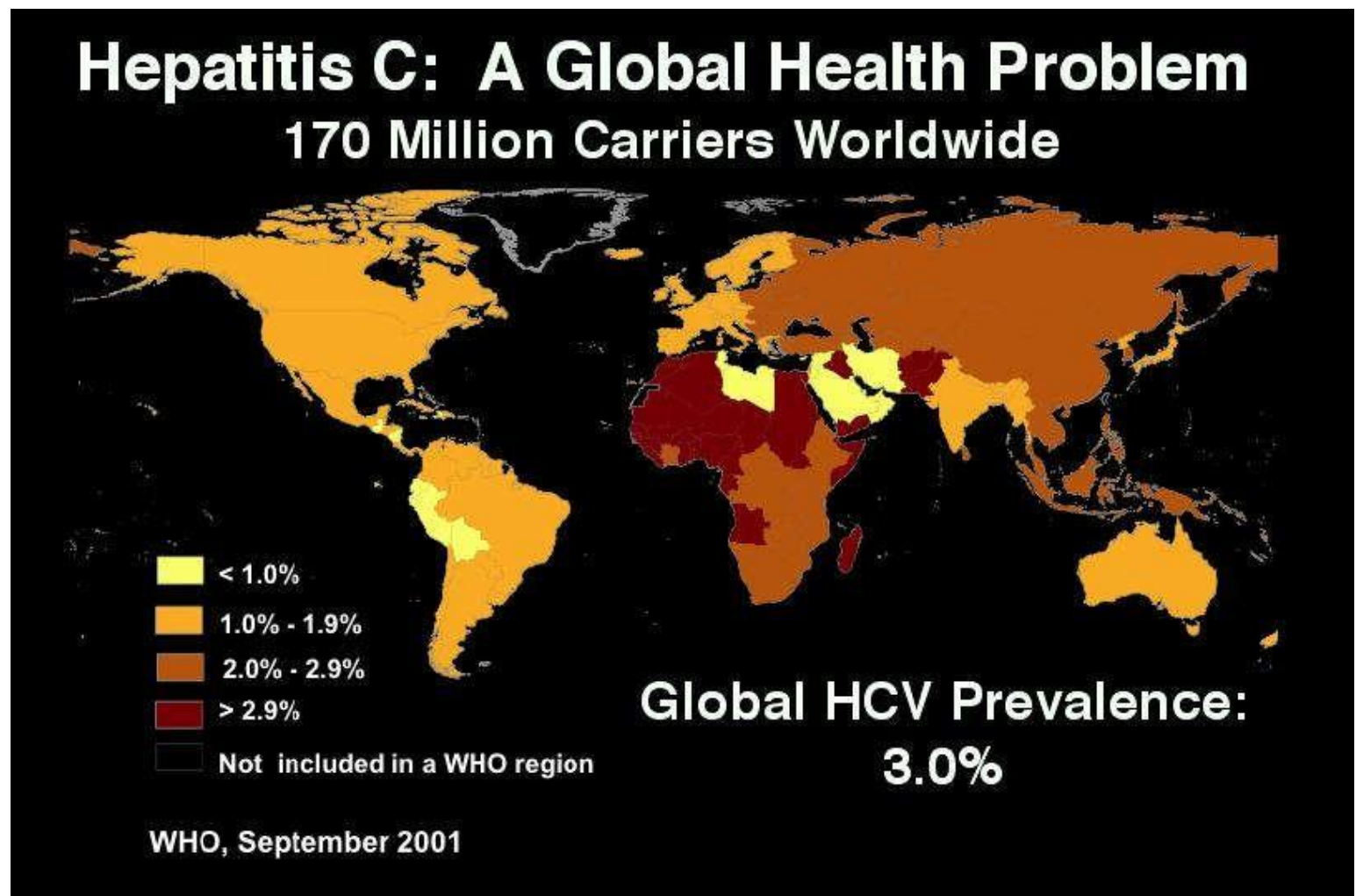
► Sexual transmission (7.1%)

► Household exposure (3.4%)

Prevalence of HCV

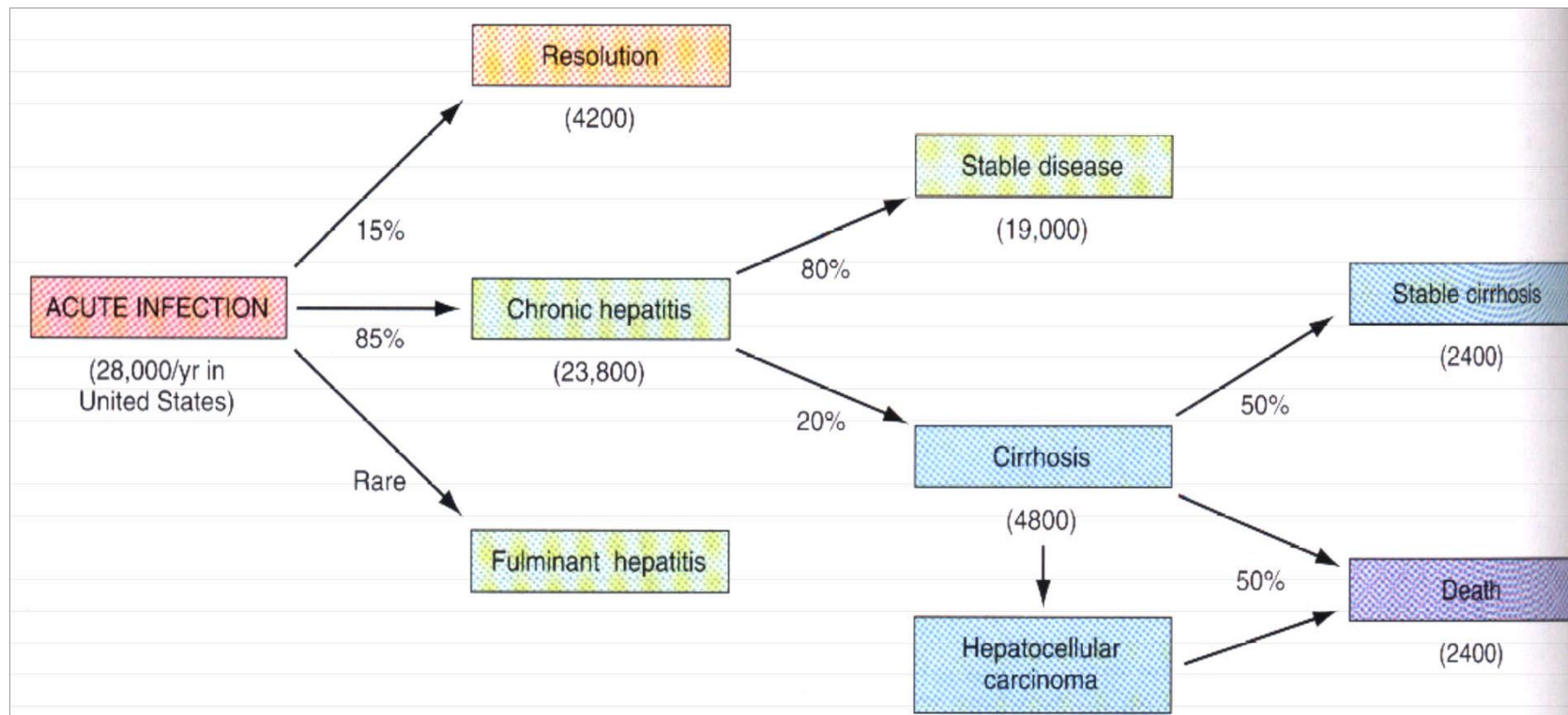
HCV is found world-wide

- In blood donors, anti-HCV antibody was found to be 0.02% in Japan, Saudi Arabia, Italy, and Spain
- Egypt → 19%
- In Western and Northern Europe, and the USA → less than 0.3%



Clinical features

- ▶ Incubation period 6-8 weeks
- ▶ Causes a milder form of acute hepatitis than does hepatitis B:
 - ▶ About $\frac{3}{4}$ of acute infections remain clinically silent
- ▶ But 50% individuals develop



Commercial HCV Assays

Indirect
Serological assays

Antibody assays

EIA-III
RIBA-III

Direct
Virological assays

HCV RNA detection

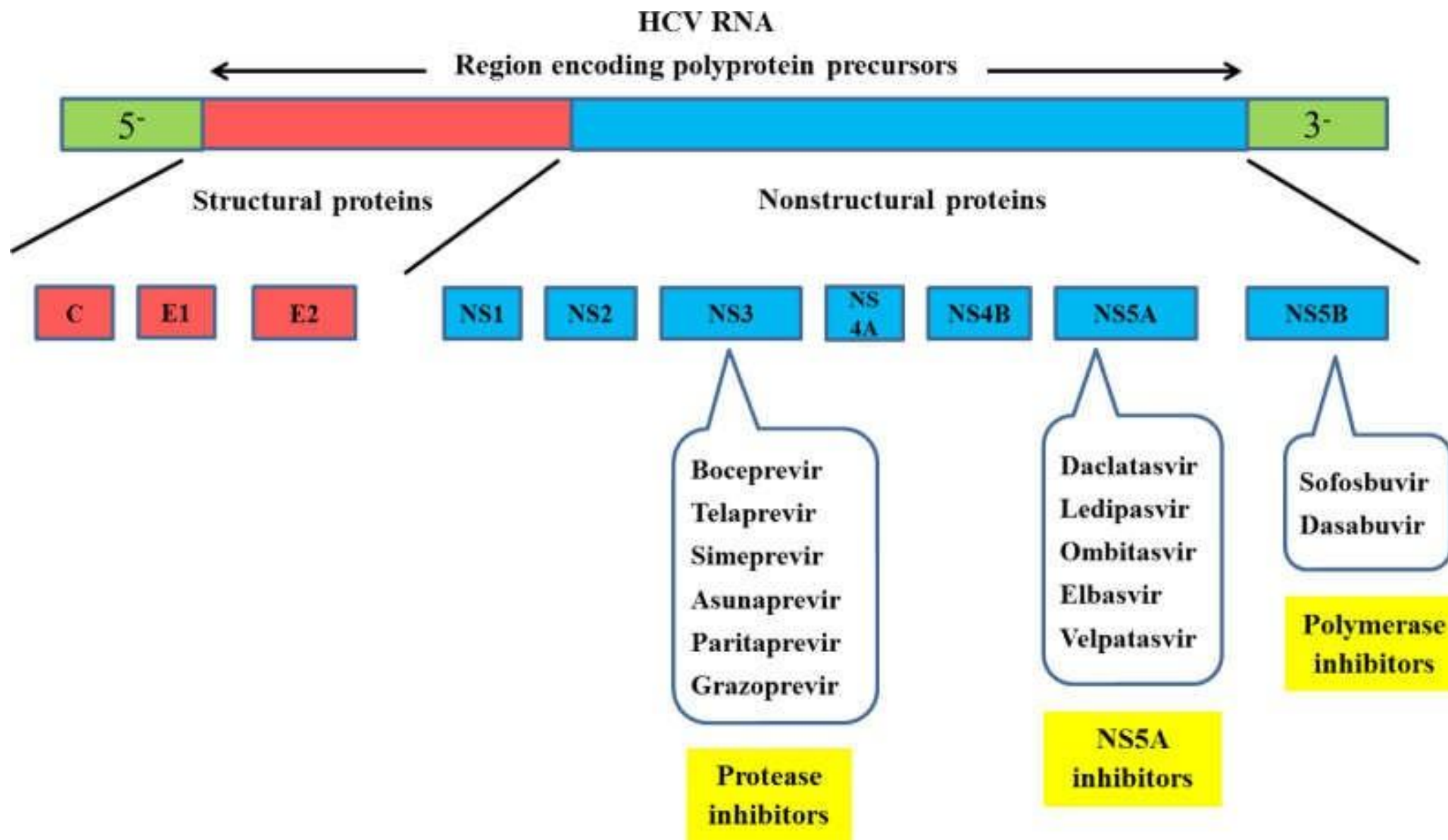
- Qualitative
- Quantitative

Molecular HCV genotyping

Therapy and prevention

- Pan-genotypic direct-acting antivirals (DAAs) for persons over the age of 12 years.
 - DAAs can cure most persons with HCV infection, and treatment duration is short (usually 12 to 24 weeks), depending on the absence or presence of cirrhosis.
- Antiviral medicines can cure more than 95% of persons with hepatitis C infection, but access to diagnosis and treatment is low.

Drug regimen	HCV genotype
Sofosbuvir/Ledipasvir +/-Ribavirin	Genotypes 1, 4, 5, and 6
Paritaprevir/Ritonavir/Ombitasvir + Dasabuvir +/- Ribavirin:	Genotype 1
Sofosbuvir + Simeprevir +/- Ribavirin:	Genotypes 1 and 4
Sofosbuvir + Daclatasvir +/- Ribavirin:	All genotypes
Paritaprevir/Ritonavir/ombitasvir +/- Ribavirin:	Genotype 4
Sofosbuvir + Ribavirin	Genotypes 2 and 3

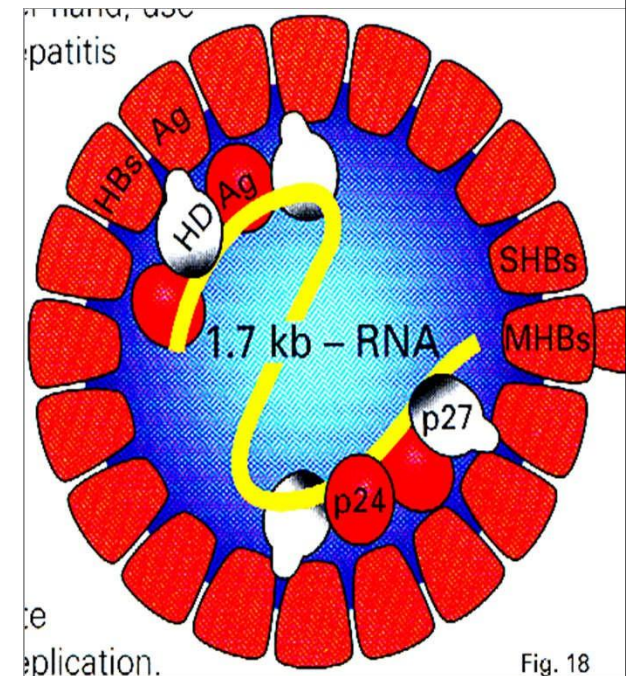


There is no vaccine for prevention of HCV

Hepatitis D Virus (HDV)- Delta virus

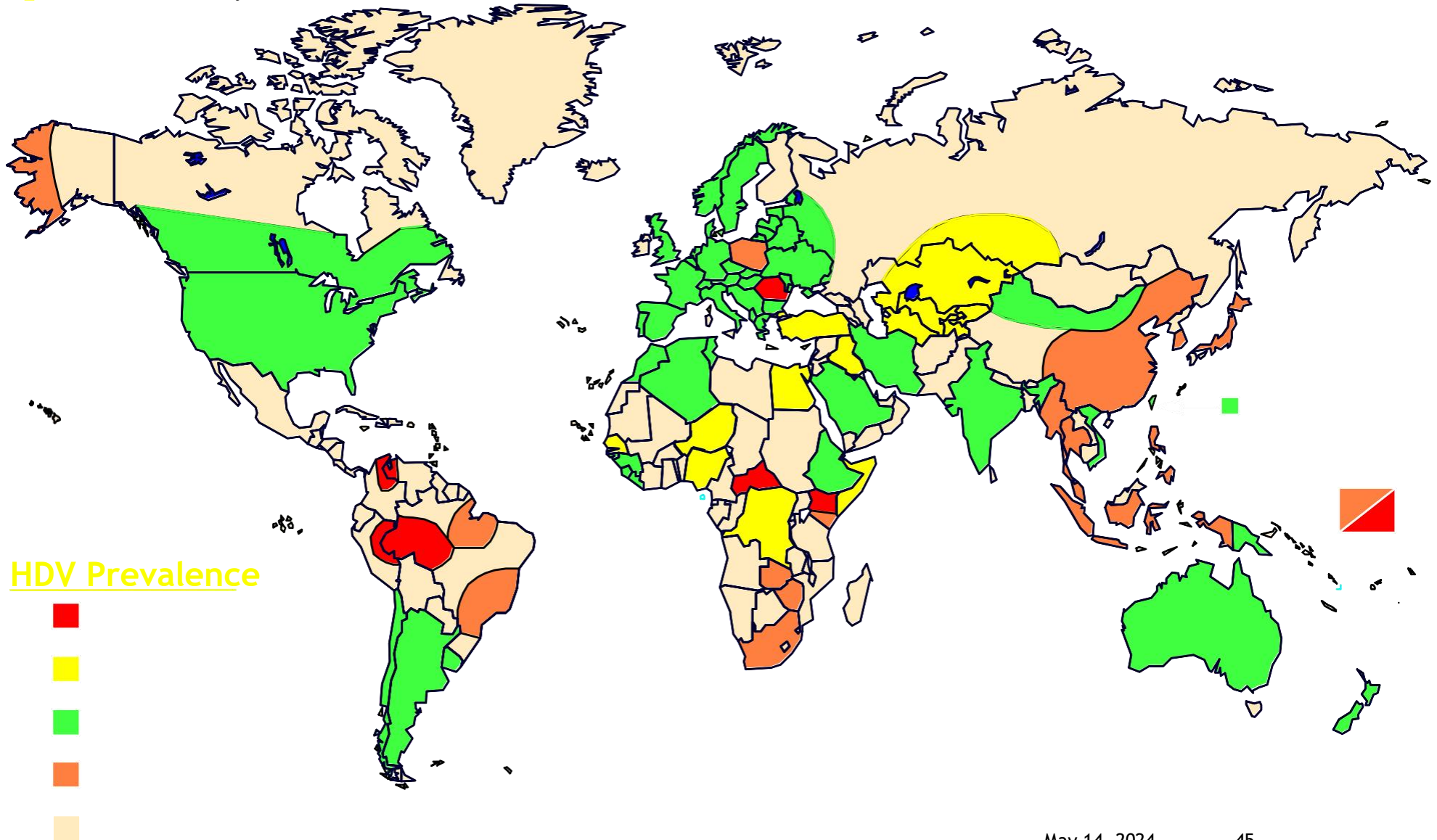
The virus

- ▶ Hepatitis delta virus is among one of the unclassified animal viruses
 - ▶ It is an *incomplete or defective virus*:
 - ▶ It *requires HBV for its replication and expression; should co-infect with HBV*
- ▶ The virion is composed of a core of HD Ag, protein, and small (1.7kb), **single stranded circular RNA genome**
- ▶ This is surrounded by an envelope containing HBsAg



Route of transmission: like HBV, parenterally

Spread: Widely distributed around the world



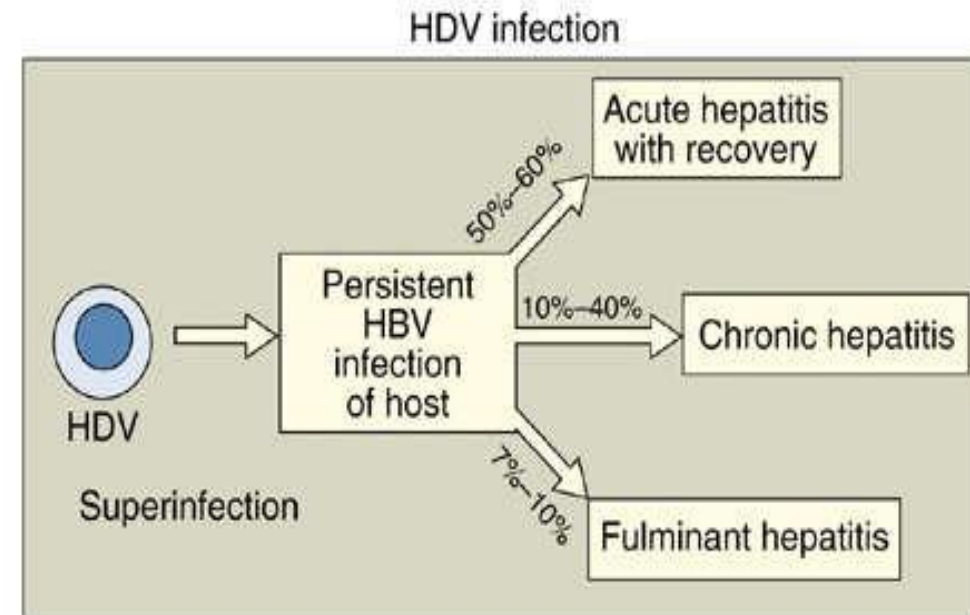
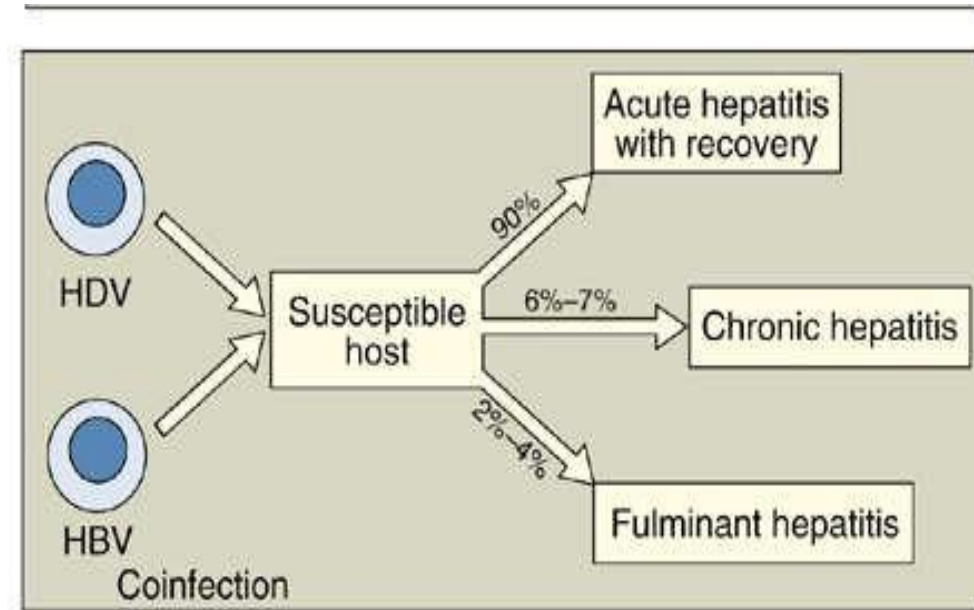
Clinical features

- ▶ Simultaneous transmission of HDV and HBV (co-infection) causes an infection clinically indistinguishable from hepatitis B
- ▶ In more than 90% of simultaneous co-infection, both agents are eliminated after the acute phase

BUT

- ▶ In HBV carriers who are superinfected with HDV, the clinical outcomes are more serious → →

Sequelae of hepatitis D virus



Diagnosis

ELISA for

- anti-HD IgG and IgM antibodies and
- Delta antigen (HDAg)

PCR for detecting HDV RNA

Therapy and immunoprophylaxis

No treatment for HDV → ===== IFN-2 α

No vaccine against HDV but vaccination against HBV can prevent HDV!

Hepatitis G virus

- ▶ Enveloped RNA virus
- ▶ Common in HCV infected patients
- ▶ Parental transmission, sexual?
- ▶ 2-39% of non A-E hepatitis
- ▶ **May delay progression of HIV disease!!!!**
- ▶ 16-43% of Fulminant hepatitis
- ▶ Detected by PCR
- ▶ No established serology

Torque teno virus (TTV)

- For the initials of the patient in whom it was isolated

- ▶ **Non-enveloped**

- ▶ **Circular, negative-sense Single strand DNA**

- ▶ **Isolated from patients post transfusion (100 %); from patients with non A-E**

- ▶ **Presents in healthy individuals (89 %) →**

- ▶ **? Normal human viral flora**

- ▶ **? Feco – oral transmission**

- TTV DNA titers were 10- to 100-fold **greater in liver tissue** than in serum, **suggesting** that it was **hepatotropic** and correlated with serum ALT elevations

- In nonhuman primates, viremia is detected 4-7 days after intravenous injection and 7-10 days after oral inoculation

Chronic Carriage: **Asymptomatic carrier state frequent**

No treatment required • **Interferon** treatment has been associated with viral clearance **during treatment of coinfections** with other hepatitis viruses.

Other hepatotropic viruses

- ▶ Acute hepatitis may also occur as part of the clinical course of a number of viral infections; that is generalized infection **plus infection of liver**

SYSTEMIC VIRUSES

Cytomegalovirus (CMV)

Epstein-Barr virus (EBV)

Herpes Simplex virus (HSV)

Varicella Zoster virus (VZV)

Adenovirus

Enteroviruses, Echovirus, Coxsackie, Rubella

Human Immunodeficiency virus (HIV)

Other (Mumps, Yellow fever, Marburg, Lassa fever, Reovirus, Rift Valley fever virus)